ModelIt!

Teacher's Guide

Can Synthetic Biology Create New Life?

Modeling Engineered Genetic Circuits and Synthetic Organism Design

Quick Reference

Grade Level	Grade 11L3
NGSS Standard	HS-LS1-1, HS-LS3-1
Time	45–60 minutes (core lesson)
Materials	Student Activity Pack, devices with ModelIt! access
Prep	Print activity packs, test ModelIt! on student devices
Grouping	Partners or small groups (2–4)

ModelIt! — Discovery Collective

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1 NGSS Standards & CAST Alignment

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Performance Expectation: HS-LS1-1, HS-LS3-1

Construct an explanation based on evidence for how the structure of DNA determines the structure of proteins which carry out the essential functions of life through systems of specialized cells. Ask questions to clarify relationships about the role of DNA and chromosomes in coding the instructions for characteristic traits passed from parents to offspring.

Three-Dimensional Learning

Science Practice	**Constructing Explanations and Designing Solutions** — Students construct explanations for how synthetic genetic circuits function and design solutions that balance functionality, safety, and evolutionary stability in engineered organisms.
Disciplinary Core Idea	**LS1.A Structure and Function / LS3.A Inheritance of Traits** — Students model how DNA sequence determines protein function in engineered genetic circuits and how synthetic genetic information is inherited and subject to mutation across generations.
Crosscutting Concept	**Stability and Change** — Students analyze how engineered biological systems maintain designed function (stability) while experiencing evolutionary pressure toward mutation (change), and how this tension determines the practical limits of synthetic biology.

CAST Testing Alignment

MULTIPLE CHOICE:

A synthetic bacterium engineered with a 12-gene metabolic pathway shows declining pollutant degradation performance over 200 generations. According to your model, which variable most likely explains this decline?

CONSTRUCTED RESPONSE:

Using evidence from your model, explain why a biocontainment system with three independent mechanisms is dramatically safer than one with a single mechanism. Calculate the approximate escape probability if each mechanism has a 10^-6 failure rate independently.

2 Scoring Rubric

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Assessment Reference

This is the same rubric students see in their Activity Pack. Use it during and after the lesson for formative and summative assessment. Circle or highlight the proficiency level for each skill.

Skill	Expert (4)	Proficient (3)	Developing (2)	Beginning (1)
Identifying Components	EXPERT Correctly identifies and sorts ALL components as internal or external with detailed reasoning	PROFICIENT Correctly identifies and sorts MOST components with some reasoning	DEVELOPING Identifies components but has difficulty sorting or explaining	BEGINNING Needs significant help identifying components
Establishing Relationships	EXPERT Accurately describes ALL positive and negative relationships with clear evidence	PROFICIENT Accurately describes MOST relationships with some evidence	DEVELOPING Identifies some relationships but struggles with direction	BEGINNING Needs help understanding how components connect
Running Simulations	EXPERT Makes accurate predictions AND explains surprising results using evidence	PROFICIENT Makes reasonable predictions and records clear observations	DEVELOPING Runs simulations but predictions or observations are incomplete	BEGINNING Needs help running simulations or recording data
Explaining Results	EXPERT Uses multiple pieces of evidence to explain HOW and WHY systems interact	PROFICIENT Uses evidence to explain system interactions	DEVELOPING Gives partial explanations with limited evidence	BEGINNING Needs help connecting evidence to explanations

Scoring Guidance

Expert (4): Student demonstrates deep understanding and can transfer knowledge to new contexts.
Proficient (3): Student meets grade-level expectations and shows solid understanding.
Developing (2): Student is making progress but needs additional scaffolding or practice.
Beginning (1): Student needs significant support; consider using Differentiation strategies (Section 7).

3 Background Content Knowledge

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For Your Confidence

This section provides the science background you need to facilitate this lesson with authority. You don't need to teach all of this — it's here so you can answer student questions accurately.

In 2010, Craig Venter's team created the first synthetic organism — a bacterium with a genome written entirely by machines and assembled from chemical building blocks. They named it Mycoplasma mycoides JCVI-syn1.0, and it became the first living cell whose parent was a computer. Since then, synthetic biology has exploded: engineered yeast now produces artemisinin (an antimalarial drug) at scale, bacteria have been programmed to detect and kill cancer cells, and the International Genetically Engineered Machine (iGEM) competition has thousands of students building biological systems from standardized genetic parts.

The DNA Programming Paradigm

Synthetic biology treats DNA as a programming language: four bases (A, T, G, C) encode instructions that cells execute. Just as software engineers write code from standardized functions, synthetic biologists assemble genetic circuits from standardized parts — promoters (on/off switches), ribosome binding sites (translation control), coding sequences (protein blueprints), and terminators (stop signals). The Registry of Standard Biological Parts catalogs thousands of characterized components. DNA synthesis technology can produce any sequence up to thousands of base pairs, and DNA assembly methods like Gibson Assembly combine multiple fragments into complete circuits in a single reaction.

Genetic Circuits as Logic Gates

Biological logic gates use molecular interactions to implement Boolean logic. An AND gate produces output only when two inputs are present — for example, a gene that activates only when both lactose AND arabinose are detected. A NOT gate inverts the signal — a repressor protein that turns off a gene when a specific molecule is present. By combining these gates, engineers build complex biological programs: a bacterium that detects a cancer biomarker AND low oxygen (indicating a tumor) then produces a cytotoxic drug. The theoretical complexity is unlimited; the practical complexity is constrained by metabolic burden and evolutionary stability.

The Evolutionary Arms Race

Every cell division is a roll of the evolutionary dice. DNA replication introduces roughly one mutation per billion base pairs per generation. For a bacterium dividing every 30 minutes, this means mutations accumulate rapidly across millions of descendants. Any mutation that reduces the metabolic cost of an engineered circuit — by disabling a non-essential gene, weakening a promoter, or breaking a regulatory connection — provides a growth advantage. Over hundreds of generations, these faster-growing mutants outcompete the engineered strain. This is not a failure of engineering; it is the fundamental nature of biology. Evolution is the strongest optimizer on Earth, and it does not care about human design intentions.

Biocontainment and Safety

Preventing engineered organisms from persisting in the environment is a critical safety requirement. Multiple biocontainment strategies exist: kill switches (genes that produce a toxin when induced, killing the cell on command), auxotrophies (dependence on nutrients like synthetic amino acids that do not exist in nature), genetic safeguards (essential genes split between a chromosome and a plasmid that is lost outside the lab), and semantic containment (recoding the organism's genetic code so that it cannot exchange genes with natural organisms). No single strategy is perfectly reliable — mutations can defeat any individual mechanism. Layering three or more independent containment systems reduces escape probability to effectively zero: if each has a 10^{-6} failure rate, three independent systems have a combined failure rate of 10^{-18}.

4 The LEVER Framework

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About LEVER

LEVER is how every ModelIt! lesson works. Students move through five phases to build, test, and refine a computational model. Each lesson is self-contained — students complete all five phases in one session.

L **Locate the System**
Students identify six key components of a synthetic biology system: circuit complexity, metabolic burden, evolutionary stability, output fidelity, growth rate, and biocontainment strength.

E **Establish Relationships**
Students map relationships showing that circuit complexity drives metabolic burden which erodes evolutionary stability, while biocontainment strength must be maintained independently through multiple mechanisms to prevent escape.

V **Visualize & Model**
Students build a computational model showing the six-component synthetic biology system with evolutionary dynamics and containment probability calculations.

E **Evaluate Outcomes**
Students run simple biosensor, complex metabolic pathway, and containment breach scenarios to discover how evolutionary pressure and circuit complexity interact to determine practical engineering limits.

R **Revise & Extend**
Students add genetic part standardization, host organism compatibility, or regulatory approval pathway to model more realistic synthetic biology development and deployment.

Dropping ModelIt! Into Your Existing Unit

ModelIt! is a **supplement**, not a replacement for your science curriculum. Each lesson stands alone, but it becomes even more powerful when students revisit their model throughout a unit. Here are easy ways to weave it in:

At the start of a unit	Use this lesson to introduce the phenomenon. Students build a “first draft” model based on what they already know. Save it.
After a reading or lab	Have students reopen their model and update it: “Based on what you just learned, would you add a component? Change a relationship?” Takes 10–15 minutes.
As a “what-if” tool	When students ask questions during class, say: “Great question — go test that in your model!” Students run new scenarios and report back.
For peer discussion	Partners compare models: “Why does your model have this arrow and mine doesn’t?” Differences spark scientific debate using evidence.
As a presentation tool	Students project their model and walk the class through it live — showing components, running scenarios, answering audience “what-if” challenges in real time.
At the end of a unit	Compare Day 1 model to final model. The growth IS the assessment — students see how their thinking evolved.

5 Slide-by-Slide Facilitation

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Slide 1: Cover

2 min

Say: *"In 2010, scientists built a living cell from scratch — its DNA written on a computer, synthesized in a machine, and inserted into a cell that booted up and started reproducing. The first organism whose parent was a computer. Today you explore what this means for the future of life itself."*

Watch For

Open with the Venter Institute story. Show the synthetic genome being assembled. Ask: if we can write the code of life, what should we write?

Slide 2: Learning Objectives

3 min

Say: *"Today you are modeling the intersection of engineering and evolution — designing living machines that biology itself is trying to reprogram."*

Watch For

Have students read objectives. Pre-teach 'genetic circuit' using the electronic circuit analogy. Pre-teach 'metabolic burden' as the cost of running synthetic programs in a living cell.

Slide 3: Big Question

3 min

Say: *"We can already write any DNA sequence we want. We can insert it into cells. We can create organisms that do things nature never imagined. But biology is not software — cells evolve, mutate, and push back against engineering. How far can we go?"*

Watch For

Quick-write: Students list potential applications of creating custom organisms and potential risks. Share out. Create two columns on the board: Promise and Peril.

Slide 4: LEVER Framework

2 min

Say: *"We are modeling a system where the most powerful force in biology — evolution — is actively working against our engineering designs. Every generation, natural selection tests whether our circuit helps or hinders the cell."*

Watch For

Preview LEVER steps. Emphasize the unique challenge: in synthetic biology, your creation can evolve to defeat your design.

Slide 5: Activity 1: Components

8 min

Say: *"Six components. Two you control: circuit complexity and biocontainment strength. Four that biology determines: metabolic burden, evolutionary stability, output fidelity, and growth rate. The cell is your factory — but it has its own priorities."*

Watch For

Guide through components. Discuss why circuit complexity and biocontainment are external — they are engineering design choices. All other properties emerge from the interaction of the synthetic circuit with the cell's biology.

Slide 6: Activity 2: Connections

8 min

Say: *"More complexity means more burden. More burden means less stability. Less stability means your circuit breaks. Evolution is the universal debugger that removes your code because it slows the cell down."*

Watch For

Help students trace the cascade from complexity through burden to stability loss. Discuss how biocontainment must be engineered independently of the functional circuit.

Slide 7: Activity 3: Simulation

10 min

Say: *"How long does your engineered function last? What happens when you push complexity? And what is the probability your organism escapes containment?"*

Watch For

Students predict evolutionary stability timelines before running simulations. Key insight: simple circuits persist, complex ones degrade. Containment breach scenario drives home the importance of redundancy.

Slide 8: Discoveries

5 min

Say: *"The most important lesson: biology is not an obedient machine. It is an evolved system with four billion years of optimization that resists reprogramming. Successful synthetic biology works WITH evolution, not against it."*

Watch For

Connect findings to real applications: artemisinin production in yeast (simple, commercially successful) versus complex therapeutic organisms (still in research). Discuss the biosafety implications.

Slide 9: STEM Challenge

12 min

Say: *"A chemical spill threatens a river ecosystem. Can you design a synthetic organism that cleans it up without becoming an ecological problem itself? The engineering starts now."*

Watch For

Groups design remediation organisms. Must address metabolic pathway, biocontainment (minimum three mechanisms), evolutionary stability strategy, and monitoring plan. Present with model evidence.

6 Answer Key & Expected Responses

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Component Sorting

- **EXTERNAL:** Circuit Complexity, Biocontainment Strength
- **INTERNAL:** Metabolic Burden, Evolutionary Stability, Output Fidelity, Growth Rate

Relationships

- **Circuit Complexity to Metabolic Burden:** POSITIVE (+) — More complex genetic circuits require more cellular resources — ribosomes to translate additional mRNA, amino acids to build additional proteins, and ATP to power the additional biochemical reactions. Each added genetic part incrementally increases the energy and material drain on the host cell.
- **Metabolic Burden to Evolutionary Stability:** NEGATIVE (-) — Higher metabolic burden creates stronger evolutionary pressure to shed the engineered circuit. Mutants that inactivate burden-imposing genes grow faster and outcompete the engineered strain. The stronger the burden, the faster the engineered function is lost through natural selection.
- **Metabolic Burden to Growth Rate:** NEGATIVE (-) — Higher metabolic burden diverts cellular resources from growth to synthetic circuit maintenance, directly slowing cell division. An engineered bacterium with a heavy synthetic circuit may divide 20-50% slower than the same strain without the circuit.
- **Biocontainment Strength to Growth Rate:** NEGATIVE (-) — Biocontainment mechanisms themselves impose metabolic costs — kill switch genes must be maintained, synthetic amino acid requirements add metabolic complexity, and auxotrophies limit growth conditions. More containment layers add incremental growth penalties.

Key Discoveries

- Synthetic genetic circuits obey the same evolutionary rules as natural genes — mutations that reduce the metabolic cost of engineered functions are positively selected, meaning evolution actively works to break engineered systems
- Circuit complexity and evolutionary stability are in fundamental tension — the more complex the engineered system, the more mutations can disable it, and the stronger the selective pressure to do so
- Biocontainment must be engineered with multiple independent safety mechanisms because any single mechanism can be defeated by mutation — redundant containment dramatically reduces escape probability
- The most successful synthetic biology applications use simple circuits with minimal metabolic burden, aligning engineered function with rather than against evolutionary pressure

7 Collaborative Learning Protocols

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Why This Matters

ModelIt! is a digital tool, but the deepest learning happens between humans. Computational modeling is most powerful when students build understanding *together* — debating relationships, challenging predictions, and constructing explanations as a community. These protocols ensure every student's voice is part of the model. Use at least ONE per lesson.

Model Gallery Walk

When: After students build models (LEVER Phase V)
Setup: Groups post their models on screens/desks. Half the class stays to "defend," half rotates with sticky notes.
Protocol: Visitors leave 2 sticky notes: one "I notice..." and one "I wonder..." After rotation, defenders read feedback aloud. Class discusses patterns.
Why it works: Students see that the SAME phenomenon can be modeled differently — and that's how real science works. (Bandura: learning through observation of peers)

Scientist Fishbowl

When: During simulation discussion (LEVER Phase E)
Setup: 4-5 students sit in center circle. Rest of class observes with clipboards.
Protocol: Inner circle discusses: "What happened in Scenario 3 and WHY?" using only evidence from their models. Outer circle records strongest arguments. Swap groups.
Why it works: Students practice scientific discourse without teacher directing. Observers learn to evaluate arguments. (Freire: dialogue as the foundation of critical consciousness)

Component Jigsaw

When: During component analysis (LEVER Phase L)
Setup: Assign each group ONE component to become "experts" on. They research it, discuss it, draw it.
Protocol: Regroup so each new team has one expert per component. Experts teach their component to the team. Team builds the model together using everyone's expertise.
Why it works: Every student is essential — the model literally cannot be built without each person's knowledge. Interdependence drives engagement. (Lave & Wenger: learning through legitimate peripheral participation)

Prediction Debate

When: Before running simulations (LEVER Phase E)
Setup: Post a scenario. Students physically move to corners: "Fire Spread will INCREASE a lot" / "INCREASE a little" / "STAY the same" / "DECREASE."
Protocol: Each corner group has 2 minutes to build their argument using model logic. One spokesperson presents. Then RUN the simulation. Losing corners explain what they missed.
Why it works: Movement + social risk + immediate feedback. Students CARE about the outcome because they committed publicly. Wrong predictions become the best learning moments.

Pair Programming for Models (Digital Collaboration)

When: Throughout model building
Setup: Two students, one device. **Driver** controls the mouse/keyboard. **Navigator** directs strategy and catches errors. Switch every 5 minutes.
Protocol: Navigator must verbalize every decision: "I think we should add a negative arrow from Rainfall to Dry Vegetation because..." Driver cannot act without Navigator's reasoning.
Why it works: Forces articulation of scientific thinking. The model becomes a shared artifact, not an individual assignment. Students who struggle with writing can demonstrate understanding verbally. Borrowed from computer science — where the best code is written in pairs.

Quick Protocol Selection Guide

Short on time (5 min)	Prediction Debate — gets bodies moving, generates energy fast
Full period available	Component Jigsaw — deep collaboration, every student contributes

After simulation day	Model Gallery Walk — peer feedback, multiple perspectives
Advanced discussion	Scientist Fishbowl — builds academic discourse skills
Every day default	Pair Programming — transforms individual work into dialogue

8 Differentiation & Extensions

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Below Grade Level

- Pre-sort components as a whole class
- Provide sentence frames for relationships
- Use visual vocabulary cards
- Pair with a strong reader

Advanced Learners

- Add 2-3 additional components
- Research real-world data
- Create multi-step causal chains
- Compare to similar systems globally

English Language Learners

- Pre-teach key vocabulary
- Provide bilingual glossary
- Use visual supports with text
- Allow verbal responses

Gifted & Talented

- Create feedback loop diagrams
- Research prevention strategies
- Write a policy brief
- Connect to climate change

9 Common Misconceptions

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Why This Matters

Addressing misconceptions proactively helps students build accurate scientific understanding. Use these as formative assessment checkpoints during the lesson.

STUDENTS MAY THINK	SCIENTIFIC UNDERSTANDING
"Synthetic biology is the same as genetic modification (GMOs)"	Traditional genetic modification typically moves existing genes between organisms — like inserting a fish antifreeze gene into a tomato. Synthetic biology goes further: it designs and builds entirely new genetic sequences from scratch, creates novel genetic circuits that implement logical programs, and can construct organisms with minimal genomes containing only the genes needed for specific functions. While GMOs modify nature's code, synthetic biology writes new code that never existed in any organism.
STUDENTS MAY THINK	**SCIENTIFIC UNDERSTANDING**
"Engineered organisms will work exactly as designed, like machines"	Biological systems are fundamentally different from machines because they evolve, respond to environmental fluctuations, and contain molecular noise that creates variability in gene expression. A genetic circuit that works perfectly in a test tube may behave differently inside a living cell due to competition for cellular resources, environmental changes, and stochastic (random) fluctuations in molecular concentrations. The model shows that even well-designed circuits produce variable outputs and degrade over evolutionary time.
STUDENTS MAY THINK	**SCIENTIFIC UNDERSTANDING**
"If a synthetic organism escapes the lab it will take over the environment"	Engineered organisms are almost always LESS fit than wild organisms in natural environments, not more. They carry metabolic burdens from synthetic circuits, lack adaptations to environmental stresses (UV radiation, predators, nutrient scarcity, temperature fluctuations), and have been optimized for laboratory conditions. A bacterium engineered to produce a chemical in a controlled bioreactor would likely be outcompeted by wild bacteria within days in a natural environment. The real biosafety concern is not dominance but unintended gene transfer to natural organisms through horizontal gene transfer — which is why genetic isolation mechanisms are included in biocontainment designs.